PACMAN Hijinks Firmware and Software (2.X)

Branches, releases and bitwise reproduction

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1 Purpose and scope

Hijinks (2.X) is the firmware and server written in 2024 for the 10-tile PACMAN 1v5 test stands at LBNL and UC Davis, between the legacy firmware (1.X, companion note) and zipline (3.X). No 2.X firmware was committed in 2024: it lived only in Vivado projects. The record is therefore the project archives, the released bundles, and two branches of `mulhearn/pacman` rebuilt from them. This note describes the branches, the releases, and how each released artifact is reproduced from the repository. Bitstreams are compared by the SHA-256 of their configuration body (first eight hex digits quoted).

2 Branches

- `hijinks-firmware` (da3f75c): firmware only, linear, on the SYNC FIX build step of the hw1v5 legacy line (3155d4a), so every commit builds by construction. Each 2.X version is one commit restored from its project archive, with the SYNC FIX build fixes applied to the archive's exported block design the same way.
- `hijinks-software` (af73a4b): the 2024 pacman-server history as committed (server 2.1 to 2.7, 2024-10-02 to 11-08), unchanged, linear.

Both stay as they are, writable for new releases. On GitHub a repository ruleset forbids force pushes and deletion on both (2026-11-08), so the history the reproductions cite cannot change. A release bundle is a pair of tags (Section 5): the firmware and the server it ships. No combined branch: both lines are linear, and the PetaLinux leg of the firmware releases is of no use (`packaging/` on `hijinks-software` holds the SYNC FIX bitstream throughout, the BOOT.BINs were assembled outside the repository).

2.1 Firmware versions

References: 2.1 the UC Davis bundle 20241005C; 2.2 the bundles 20241007B and 20241010A (two independent bundles, the same bitstream); 2.7 its project archive's own build (no bundle); 2.8 the LBNL bundle 20241026A; 2.9 the LBNL bundle 20241029A; 2.10 its release 20261020A. The project archives are `pacman-fw-2v1.xpr.zip` to `pacman-fw-2v9.xpr.zip` (2v1, 2v2, 2v7, 2v8, 2v9).

3 Releases

Bundle	Kind	FW	Server	Notes
UCD 1v5_20241005A_ucd_syncfix	legacy	SYNC FIX	I2C 5.2	the branching-off point for hijinks

Bundle	Kind	FW	Server	Notes
UCD 1v5_20241005B_ucd_devhiijinks	development	SYNC FIX	2.1	server only
UCD 1v5_20241005C_ucd_devhiijinks	development	2.1	2.1	
UCD 1v5_20241007A_ucd_devhijinks	development	2.1	2.2	server only
UCD 1v5_20241007B_ucd_hiijinks	released	2.2	2.2	first hijinks bundle (not successfully deployed)
LBNL 1v5_20241010A_hijinks	released	2.2	2.4	first deployed
LBNL 1v5_20241026A_hiijinks	released	2.8	2.4	
LBNL 1v5_20241029A_hijinks	unreleased	2.9	—	experimental; BOOT.BIN and xsa only
server 2v5, 2v7	software only	(2.8)	2.5, 2.7	tarballs; 2.7 is the CERN deployment
LBNL 1v5_20261020A_hijinks	released	2.10	2.7	released privately for CERN

The released directory names keep their "hiijinks" spelling. Every bundle's `image.ub` is the same (b64633c2). The rootfs images were checked by the server binaries they carry (`pacman_cmdserver`, `pacman_dataserver`): each is exactly a released server tarball's.

4 Reproducing the server

The server is a PetaLinux 2018.3 application. Its recipe, `pacman-server.bb`, is in `petalinux/pkg/pacman-server`; the branch's README builds it with `petalinux/scripts/build_pacman_server.sh` (`petalinux-build -c pacman-server`), and it ships as the files of the resulting RPM. The binaries are stripped and carry no build dates. Every released server was reproduced bitwise from its commit (2026-10-31, Table 3).

A plain build differs from the release in two ways, both from the build machine rather than the source:

- **Link order.** The Makefile collects its sources with `$(wildcard src/*.cc)`, whose order is the directory order of the build machine's file system. The 2024 builds linked `pacman_simulated.o`, `message_format.o`, `chan_map.o` (from 2.2 on), `rx_buffer.o`, `pacman_i2c.o`, `dma.o`, `tx_buffer.o`, `pacman.o`, then the application. The order was found by locating each object's code in the released binary, and confirmed by the 2024 build log.
- **Build path.** The build ID (20 bytes) and the `.gnu_debuglink` checksum (4 bytes) are hashes over the debug information, which records the absolute path of the build's sysroot. The 2024 builds were made in `/home/mulhearn/research/petalinux-only/pacman/petalinux/ramdisk` on vonnegut.

Each release was therefore built from its commit's recipe with the objects kept, then relinked inside the PetaLinux 2018.3 container in the release's order, with the work directory mounted at the 2024 path, followed by Yocto's packaging steps (`debugedit`, `objcopy --only-keep-debug`, `--add-gnu-debuglink`, `strip`). The result equals the release in every byte. The release commits of 2.2 and 2.5, each of which spans several commits, were pinned by the tarballs' file times (a minute or two after 85436ea and ea74a01) and confirmed by the match. The servers in the bundles' rootfs images are the tarballs', so every released hijinks server is covered.

5 Tags

Every released firmware and server carries an annotated tag (2026-11-08), read-only on GitHub through a second ruleset. Each annotation gives the commit, the build recipe, the result (bitwise against the release), the release bundles that used it, and for the firmware the SHA-256 of the bitstream and hardware

Version	Commit	Date	Body	Content
—	402413b			firmware only: removes the PetaLinux project and server (now on the software branch), `setup.sh`, and the bare-metal apps needing an AXI GPIO the design lacks
2.1	b163935	2024-10-05	f87eb690	10 tiles, renamed I/O ports, version registers `0x0`–`0xC`, 10-bit masks
2.2	d70db3c	2024-10-07	1431d8cb	hijinks constraints, 32 UARTs across 10 tiles
2.7	96d41a0	2024-10-16	6ba3a82a	UART TX inter-word pause, RX mask (2.3–2.6 were unreleased test increments)
2.8	4b463da	2024-10-26	c4a85eb2	sync/reset on the trigger line too, 32-bit RX mask, 160 µs TX gap
2.9	e250cfd	2024-10-29	0be6948d	sync reset pulse 15 cycles (`C_SYNC_RST_CYCLES` 10 → 15)
2.10	da3f75c	2026-10-21	ede22a64	adjustable MCLK (v1.3-1.0's) on 2.8, for CERN

Table 1: The firmware commits. 2.1–2.9 were rebuilt from the repository (Vivado 2020.2, `recreate_xpr.tcl` then `build_xpr.tcl`, about 90 minutes each) and are bitwise identical to their references (2026-10-13); the commits have since been replayed with the same trees, and each is built once more from a fresh clone of the repository (bitwise: 2.1, 2.2, 2.7, 2.8, 2.9, 2026-11-01; 2.10, 2026-11-02). 2.10 was built from this tree when released.

Server	Commit	`pacman_cmdserver`	`pacman_dataserver`	Shipped in
2.1	dcdbc71	27f457d0	19b33c8c	tarball 2v1 (= 05B's); rootfs of 05C
2.2	85436ea	af9df9ab	f0c56599	tarball 2v2; rootfs of 07A, 07B
2.4	2f6f6bf	629674af	230cd1df	tarball 2v4; rootfs of 10A, 26A
2.5	ea74a01	8e9c70b8	665e4383	tarball 2v5
2.7	af73a4b	61f8cb40	13fcbb44	tarball 2v7

Table 3: The released server binaries (SHA-256, first eight hex digits), each reproduced bitwise from its commit on `hijinks-software`.

description. Every firmware version was built once more from a fresh clone of the repository (2026-11-01 and 02), bitwise.

Firmware tag	Commit	Body	Server tag	Commit	Link order
`hijinks/hw1v5/fw-2.1`	b163935	f87eb690	`hijinks/hw1v5/server-2v1`	dcdbc71	without chan_map
`hijinks/hw1v5/fw-2.2`	d70db3c	1431d8cb	`hijinks/hw1v5/server-2v2`	85436ea	with chan_map
`hijinks/hw1v5/fw-2.7`	96d41a0	6ba3a82a	`hijinks/hw1v5/server-2v4`	2f6f6bf	with chan_map
`hijinks/hw1v5/fw-2.8`	4b463da	c4a85eb2	`hijinks/hw1v5/server-2v5`	ea74a01	with chan_map
`hijinks/hw1v5/fw-2.9`	e250cfd	0be6948d	`hijinks/hw1v5/server-2v7`	af73a4b	with chan_map
`hijinks/hw1v5/fw-2.10`	da3f75c	ede22a64			

Table 4: The hijinks tags.

```
git tag -l 'hijinks/*'                                  # every hijinks tag
git tag -l --format='%(contents)' hijinks/hw1v5/fw-2.8  # one tag's annotation
git checkout hijinks/hw1v5/server-2v7                   # the state, ready to build
```

The bundle table (each bundle's firmware tag, server tag and artifact hashes) is also kept with the artifacts, as `HIJINKS.md` in `mulhearn/pacman_releases`. A new release adds a tag and a row in both.

6 Still open

- Validation of 2.10 with the hardware loopback card at 5 and 10 MHz (tabled).